Quick Ligation™ Kit Protocol (NEB #M2200)

Materials Required but not Supplied

Quick Ligation™ Kit

- Nuclease-free Water (NEB #B1500)

Overview

The Quick Ligation Kit enables high-throughput cloning into vectors, library construction, T/A cloning, linker ligation, and re-circularization of linear DNA. Use this rapid and highly efficient protocol to ligate single-base overhang, cohesive end, or blunt end DNA fragments in 5 minutes at room temperature (25°C). For efficient blunt-end ligation into DNA cloning vectors, begin with the compatible Quick Blunting™ Kit (NEB #E1201) protocol, which converts up to 5 µg of DNA with incompatible 5´or 3´overhangs to 5´phosphorylated, blunt-ended DNA per reaction. Ensure successful transformations downstream in all cloning scenarios by using the protocols supplied with NEB's compatible Cloning Competent Cell Strains which are available in a variety of formats.

Precautions:

- Thaw and resuspend Quick Ligase Reaction Buffer at room temperature before use.
- Transformation efficiency can drop, particularly if the ligation reaction is carried out overnight.

Protocol

1. Set up the following reaction in a microcentrifuge tube on ice. (Please note that the enzyme should always be added last.)

   Use NEBiocalculator® to calculate molar ratios.

   NEBioCalculator

COMPONENTS	20 µl REACTION	FINAL CONC. OR AMOUNT
Quick Ligase Reaction Buffer (2X)*	10 µl	1X
Vector DNA (4 kb)	X µl	50 ng (0.020 pmol)**
Insert DNA (1 kb)	X µl	37.5 ng (0.060 pmol)**
Quick Ligase	1 µl	
Nuclease-free Water	to 20 µl	

**Quick Ligase Reaction Buffer should be thawed and resuspended at room temperature.*

***Recommendations are based on a standard ligation reaction in a cloning workflow. Molar ratios of the two DNA fragments being ligated (vector and insert) can be adjusted based on the relative size differential. In the example shown above, a 1:3 vector-to-insert ratio has been used.*

2. Gently mix the reaction by pipetting up and down and microfuge briefly.

3. Incubate at room temperature (25°C) for 5 minutes.

4. Chill on ice and transform 1-5 µl of the reaction into 50 µl competent cells. Alternatively, Store at -20°C.

5. Transform the ligation product into *E. coli* cells following the protocol included with the selected NEB competent cell strain and format.

General Guidelines

1. **Cells:**
   Competent cells can vary by several logs in their competence. Perceived ligation efficiency directly correlates to the competence of the cells used for transformation. Always transform uncut vector as a control for comparison purposes. A spin column is required to remove PEG from the Quick Ligation Kit prior to electroporation.

2. **Reaction Time:**
   Most ligations performed using the Quick Ligation™ Kit reach an endpoint at 5 minutes or less at 25°C. Incubation beyond this time provides no additional benefit. In fact, transformation efficiency starts to decrease after 1 hour and is reduced by up to 75% if the reaction is allowed to go overnight at 25°C.

3. **DNA:**
   Purified DNA for ligations can be dissolved in nuclease-free water; TE or other dilute buffers also work well.
   For optimum ligation, the volume of DNA and insert should be 10 µl before adding 2X Quick Ligation Buffer. For DNA volumes greater than 10 µl, increase the volume of 2X Quick Ligation Buffer such that it remains 50% of the reaction and correspondingly increase the volume of ligase.

   We recommend a vector amount of 20-30 fmol with an overall concentration of vector + insert between 1-10 ng/µl for efficient ligation. Concentration lower than 1 ng/µL may result in intramolecular ligation/circularization of the fragments. Vector: Insert molar ratios between 1:1 and 1:10 are optimal for single insertions (up to 1:20 for short adaptors). Use NEBiocalculator to calculate molar ratios.

4. **Biological Constraints:**
   Some DNA structures, including inverted and tandem repeats, are selected against by *E. coli*. Some recombinant proteins are not well tolerated by *E. coli* and can result in poor transformation or small colonies.

Related Resources

- Fast Cloning: Accelerate your cloning workflows with reagents from NEB

- DNA Ligase Selection Chart
- PCR & DNA Cleanup
- Competent Cell Selection Guide
- NEBioCalculator®